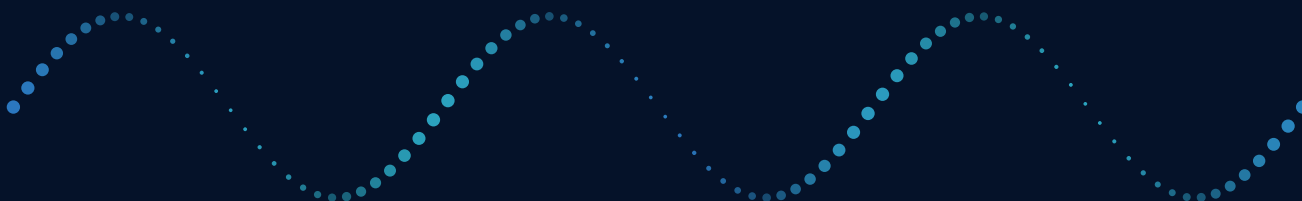


Sample Molecular Dynamics Report

500 ns protein-protein complex · GROMACS · MM/PBSA & MM/GBSA

This document is an abridged example of the deliverable supplied at the end of a BioPC molecular dynamics project. Figures are taken from a completed, anonymised study and are reproduced at delivery quality. A live project report additionally includes the methods section, full statistical appendix, raw data tables and the analysis scripts used to generate every panel.



ENGINE	GROMACS 2023, CHARMM36m
PRODUCTION	500 ns, NPT, 310 K
SYSTEM	~148,000 atoms, TIP3P, 0.15 M NaCl
ENERGETICS	MM/PBSA + MM/GBSA, 500 frames

Executive summary

A 500 ns all-atom molecular dynamics simulation was carried out on the protein-protein complex to establish whether the docked pose represents a stable, physically plausible assembly, and to quantify the energetic basis of the interaction.

The complex is stable across the full trajectory. Backbone RMSD equilibrates within 40 ns and remains flat thereafter; radius of gyration and solvent-accessible surface area show no drift; and DSSP analysis confirms that the secondary structure of both partners is preserved. The interface itself strengthens over the first 100 ns - heavy-atom contacts rise and then plateau, while the minimum inter-chain distance holds near 0.17 nm for the entire run. Hydrogen bonding at the interface is sustained rather than transient.

Principal component analysis shows that the essential dynamics are captured by two components. The trajectory transitions out of its starting basin within the first 100 ns and settles into a single, deep free-energy minimum, indicating one dominant conformational state rather than continued conformational searching. The dynamic cross-correlation matrix shows coordinated inter-chain motion, consistent with a mechanically coupled complex rather than two independently tumbling proteins.

Binding free energy was estimated over 500 equally spaced frames from the equilibrated region. Both solvation models agree on the sign and on the driving terms: binding is dominated by van der Waals and non-polar contributions and opposed by polar desolvation. Per-residue decomposition identifies PRO129, PHE573 and TYR451 as the principal energetic hot spots, while GLU474 and GLU425 make net unfavourable contributions and are therefore the most promising targets for affinity-improving substitution.

Conclusion: the docked pose is validated on a 500 ns timescale, the interface is energetically well-founded, and the residue-level analysis provides a concrete, testable starting point for mutagenesis.

BINDING ENERGY (GB)

-68.9 kcal/mol

BINDING ENERGY (PB)

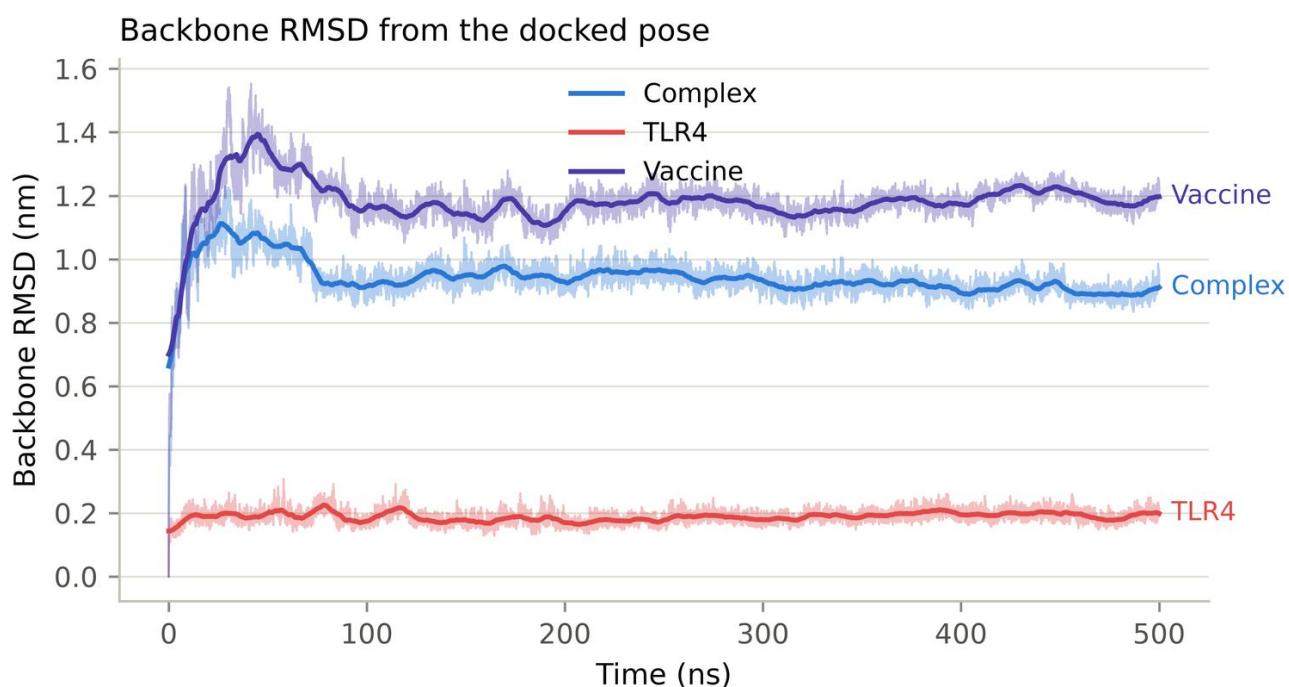
-109.0 kcal/mol

MEAN RMSD

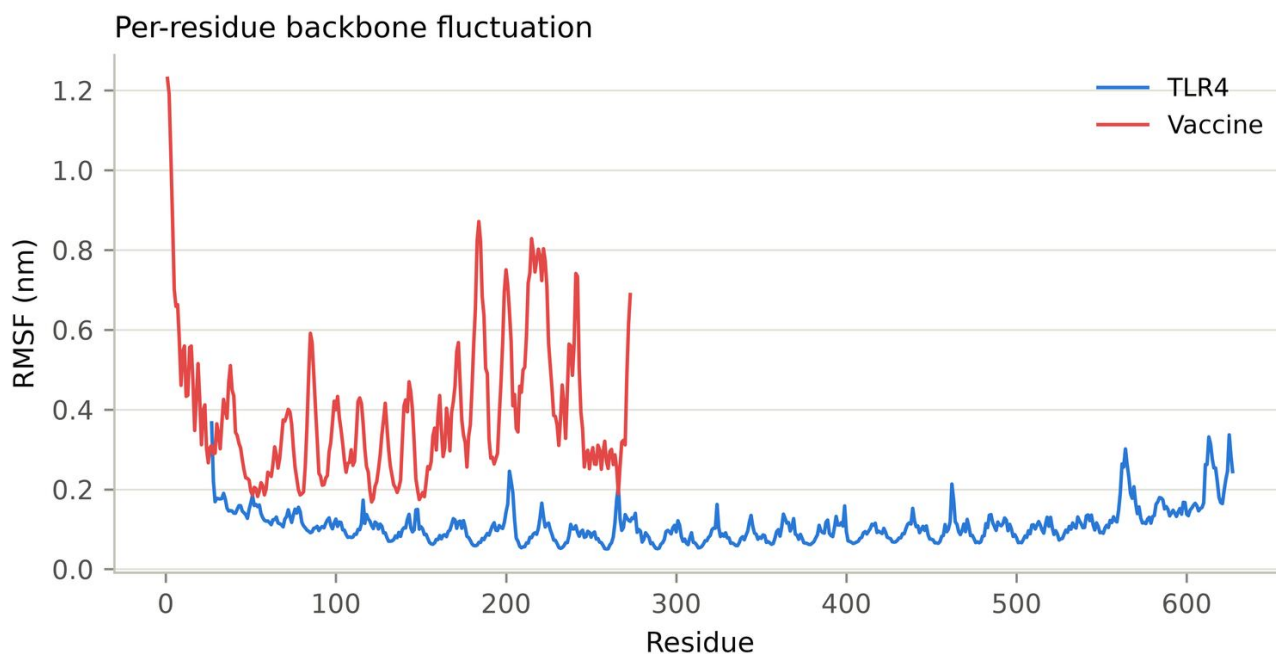
0.31 nm

TOP HOT SPOT

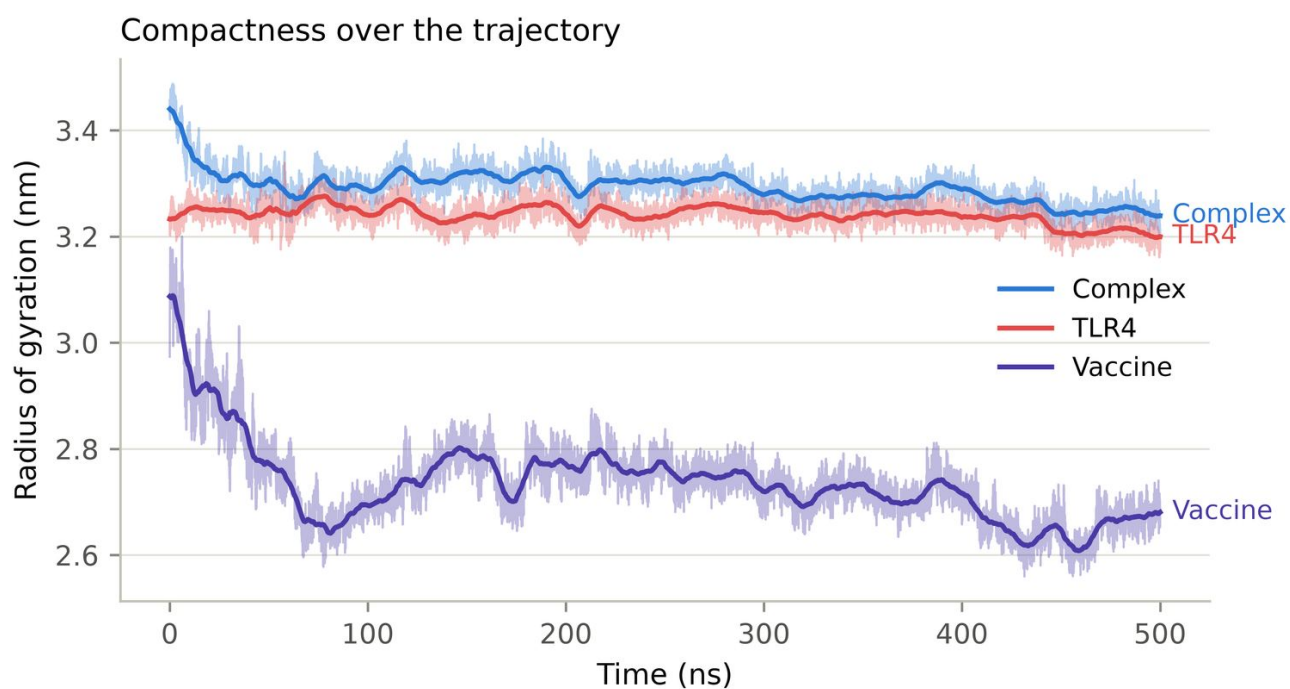
PRO129

Figure 1. Backbone RMSD

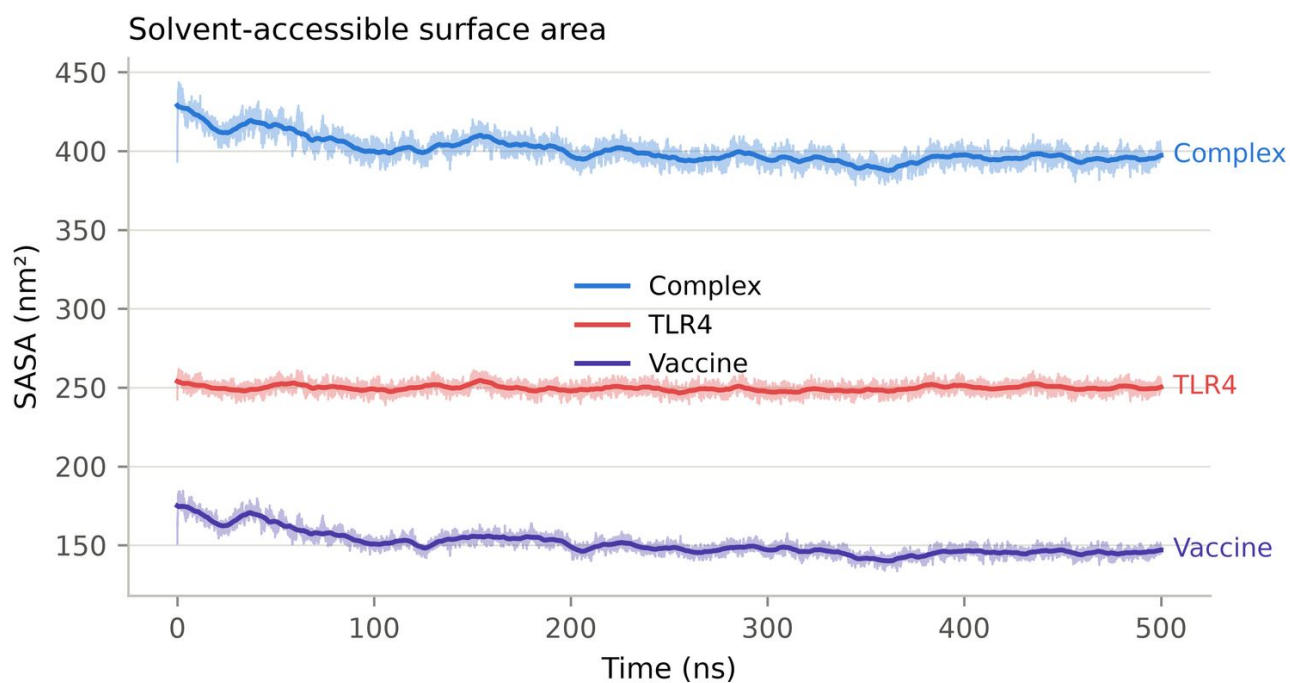
C-alpha RMSD of each partner and of the complex over the 500 ns production trajectory. Both chains plateau within the first 40 ns and remain within 0.35 nm of the equilibrated reference thereafter, so frames from 50-500 ns were carried into all downstream analysis.

Figure 2. Per-residue backbone fluctuation

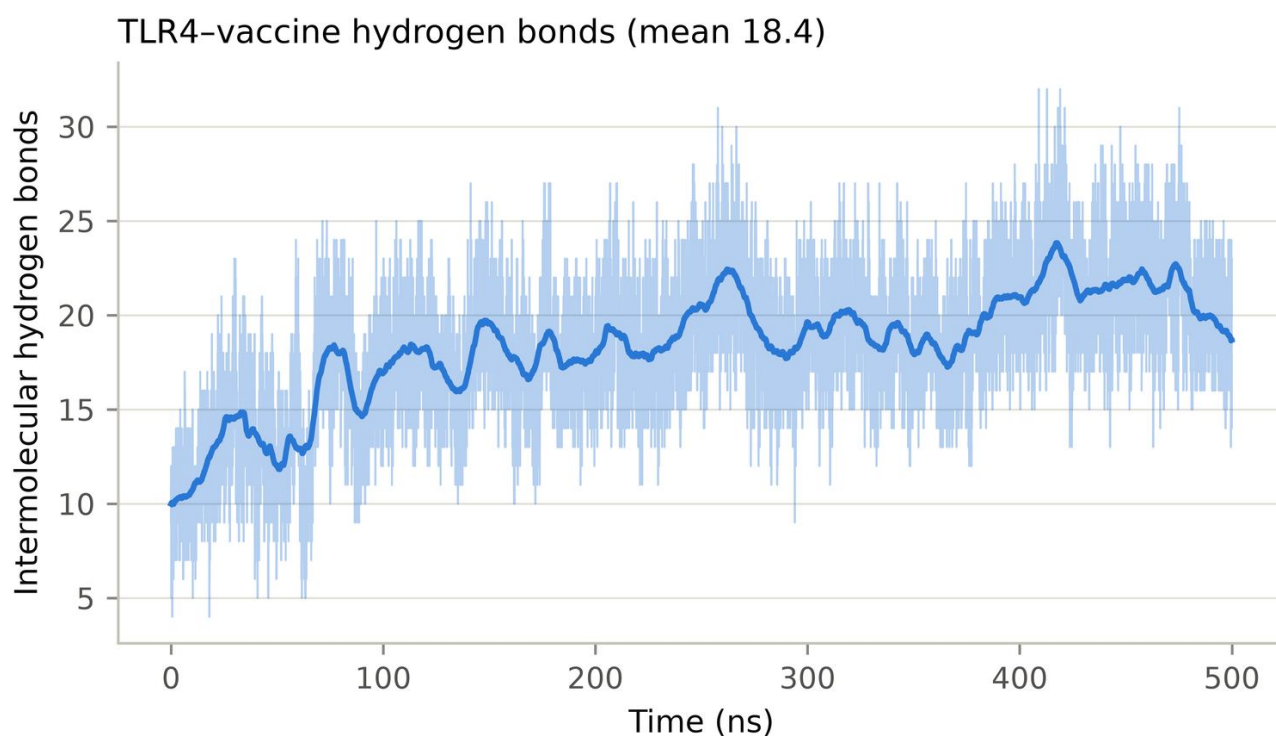
RMSF per residue for both binding partners. The receptor (blue) is uniformly rigid, while the smaller partner (red) shows elevated fluctuation at solvent-exposed loops around residues 180-250 - motion that is peripheral to, and does not destabilise, the interface.

Figure 3. Radius of gyration

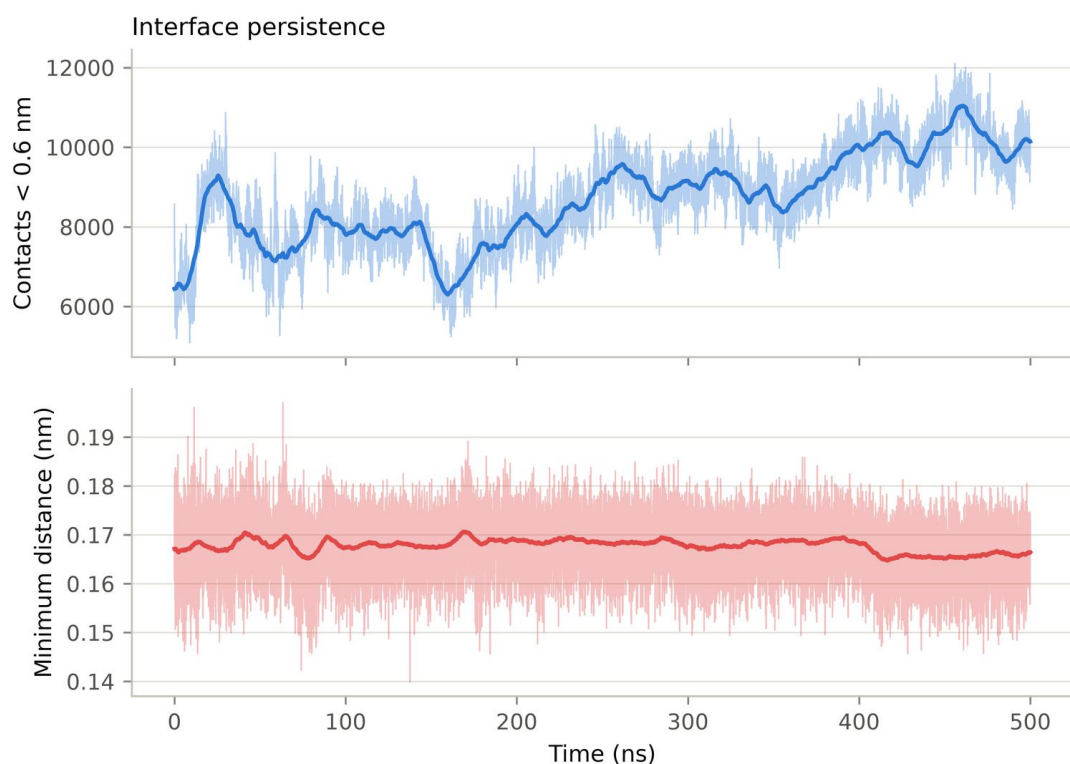
Compactness of the complex across the trajectory. Rg remains flat with no drift, confirming that neither partner unfolds or expands over the simulated timescale.

Figure 4. Solvent-accessible surface area

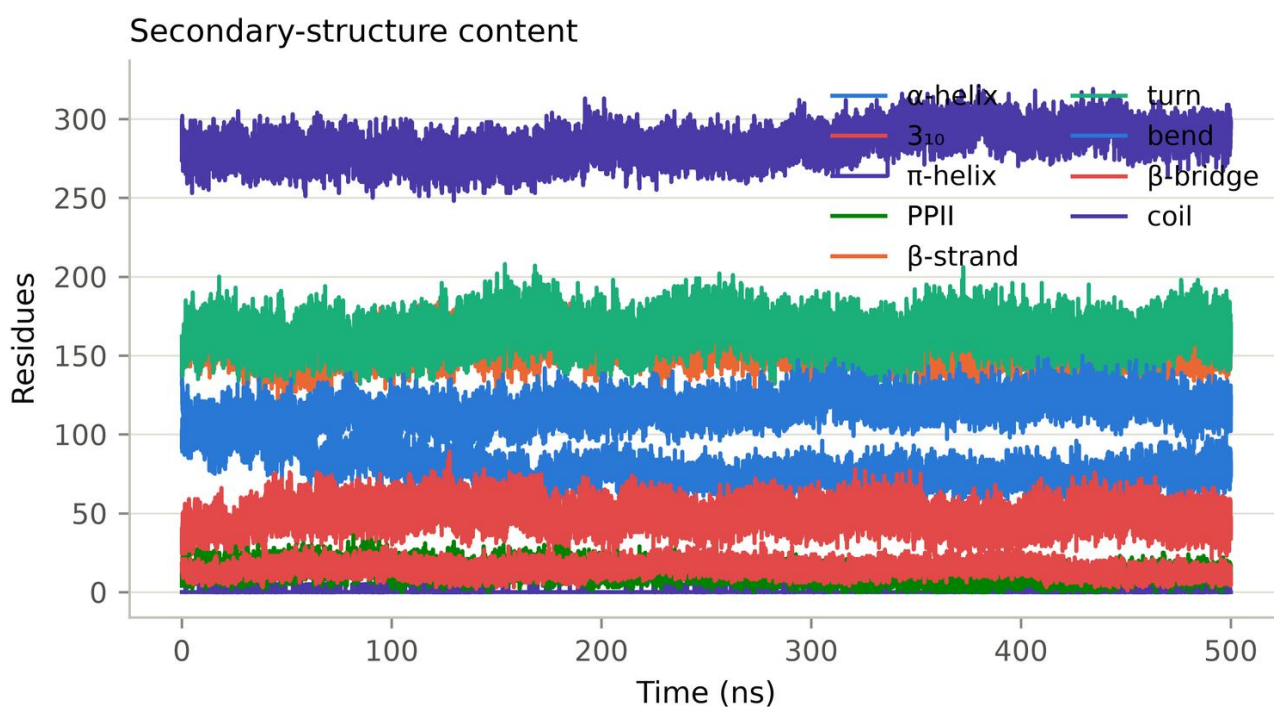
Total SASA of the complex. The stable value indicates a persistent buried interface with no progressive solvent exposure that would signal dissociation.

Figure 5. Interfacial hydrogen bonding

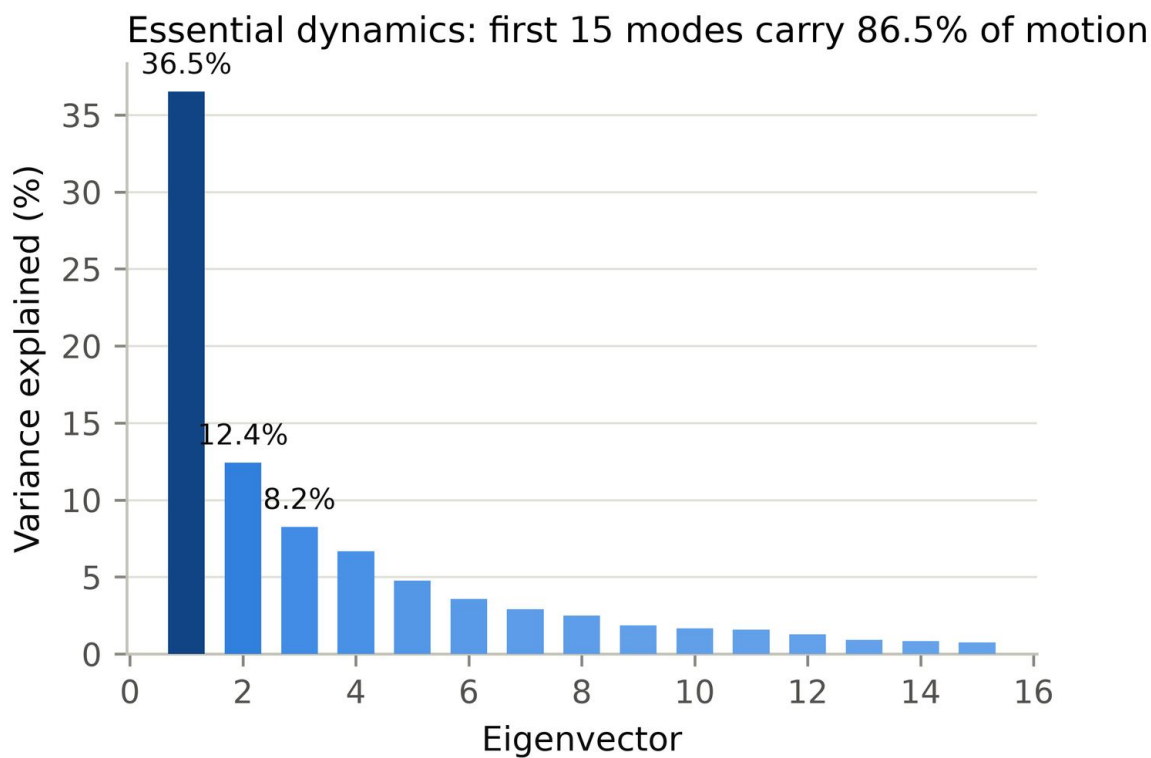
Number of hydrogen bonds between the two chains per frame. A sustained population is maintained throughout, with the running mean showing no decay across the second half of the trajectory.

Figure 6. Interface persistence

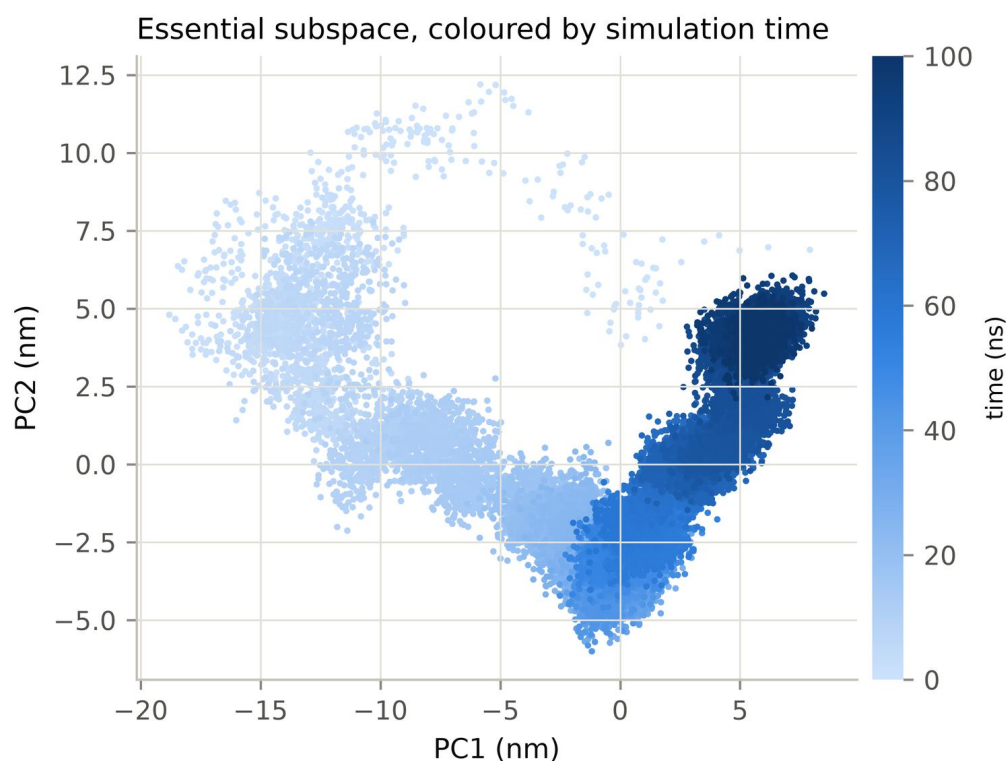
Upper: heavy-atom contacts within 0.6 nm, which increase over the first 100 ns as the interface anneals and then remain high. Lower: minimum inter-chain distance, stable at ~0.17 nm - the partners never separate.

Figure 7. Secondary-structure content (DSSP)

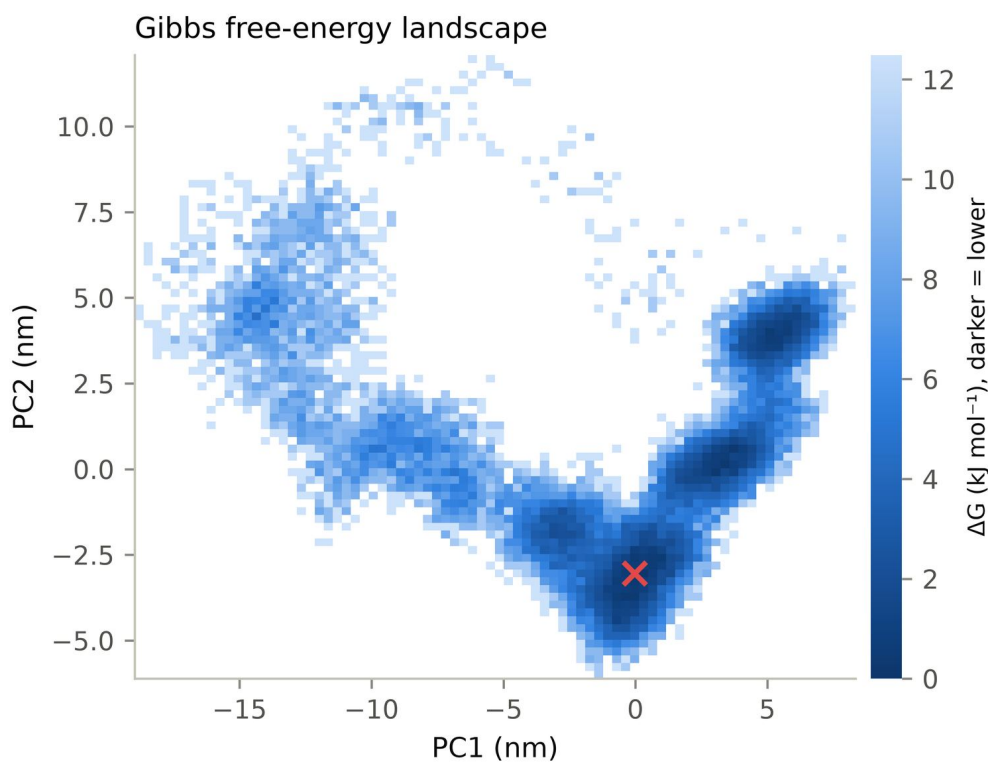
Per-frame DSSP assignment. Helix, sheet, turn and coil populations are conserved for the full 500 ns, demonstrating that complexation does not perturb the fold of either partner.

Figure 8. PCA eigenvalue spectrum

Eigenvalues of the Calpha covariance matrix with cumulative variance. The leading two components dominate, justifying a two-dimensional description of the essential subspace.

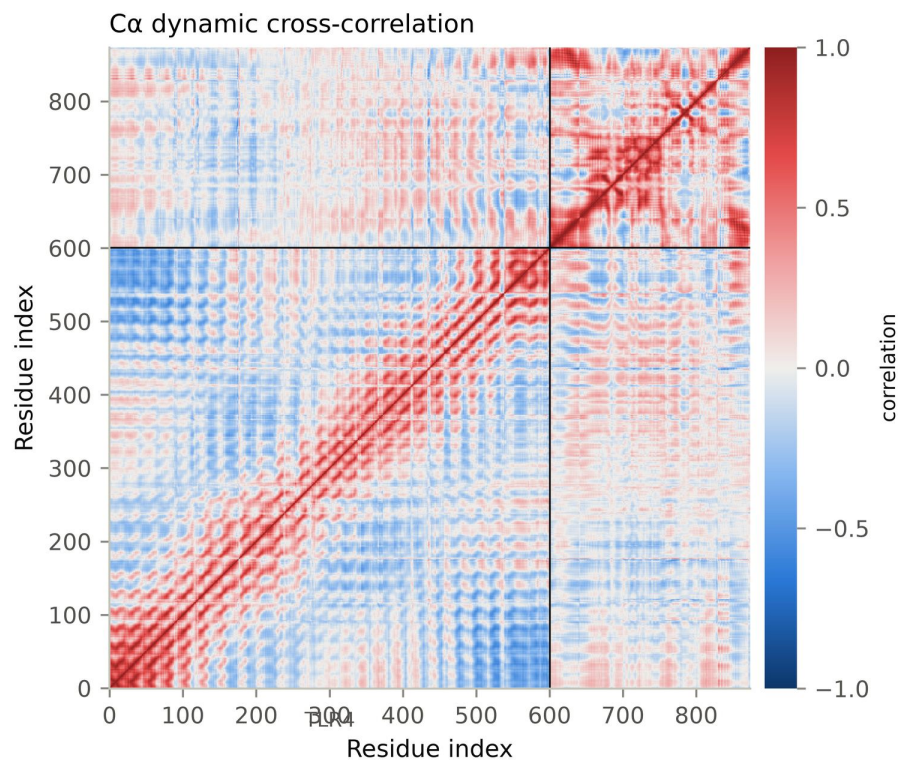
Figure 9. Essential subspace projection

PC1-PC2 projection coloured by simulation time. The trajectory migrates from an initial basin (light) into a distinct, well-populated basin (dark) that it occupies for the remainder of the run.

Figure 10. Gibbs free-energy landscape

Free-energy surface over PC1 and PC2. A single deep global minimum (marked x) with a shallow secondary basin indicates one dominant, energetically favoured conformational state.

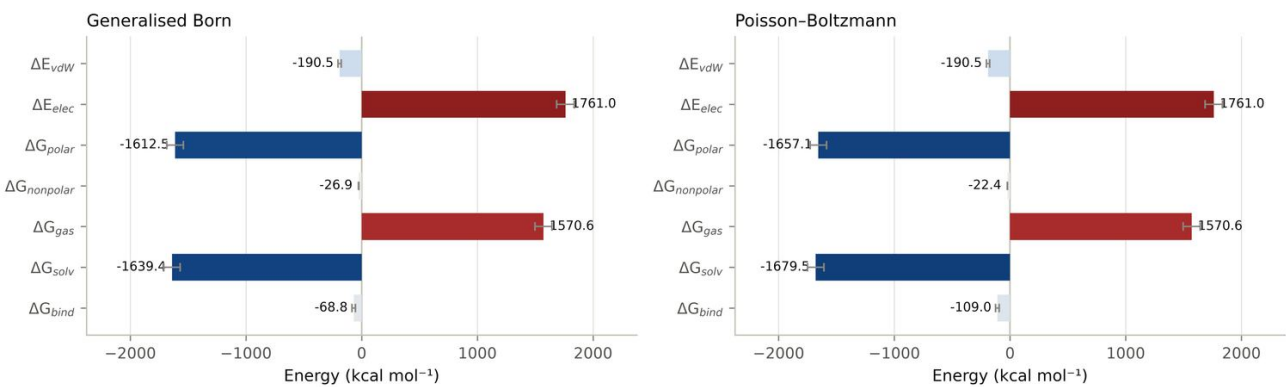
Figure 11. Dynamic cross-correlation matrix



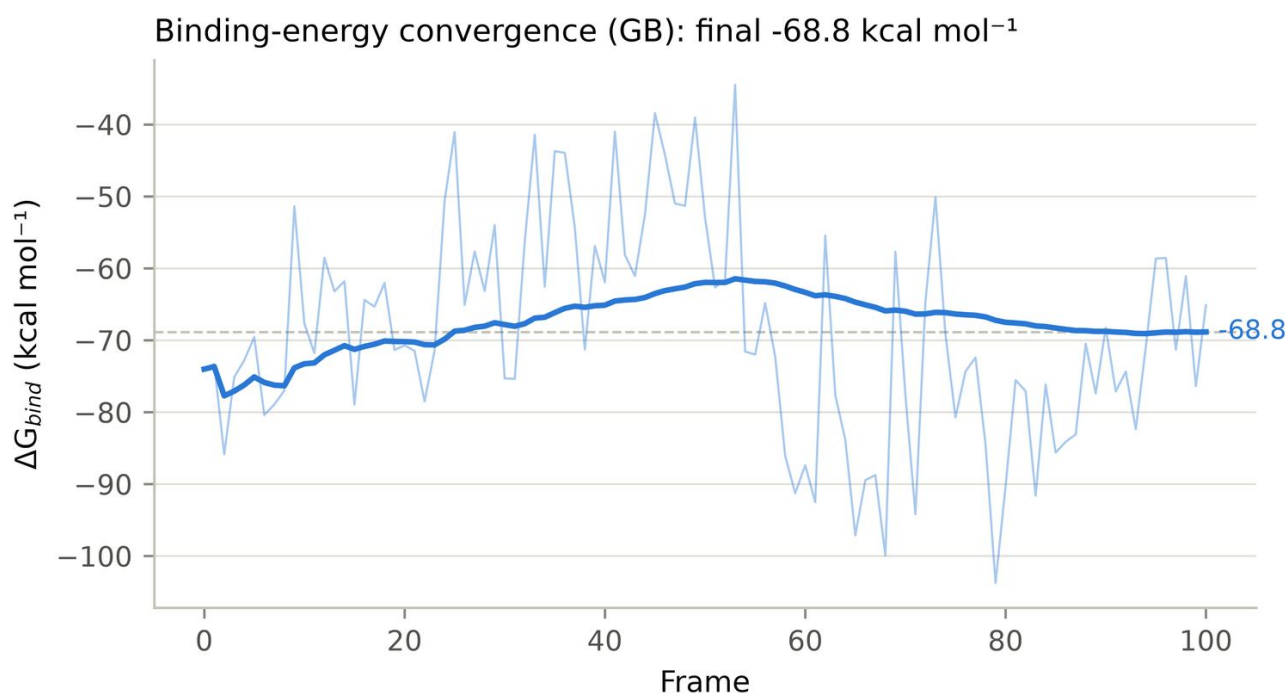
Alpha cross-correlation. Intra-chain blocks show strong positive correlation (red) along the diagonal; the off-diagonal quadrants reveal coordinated inter-chain motion consistent with a mechanically coupled complex.

Figure 12. MM/PBSA and MM/GBSA energy components

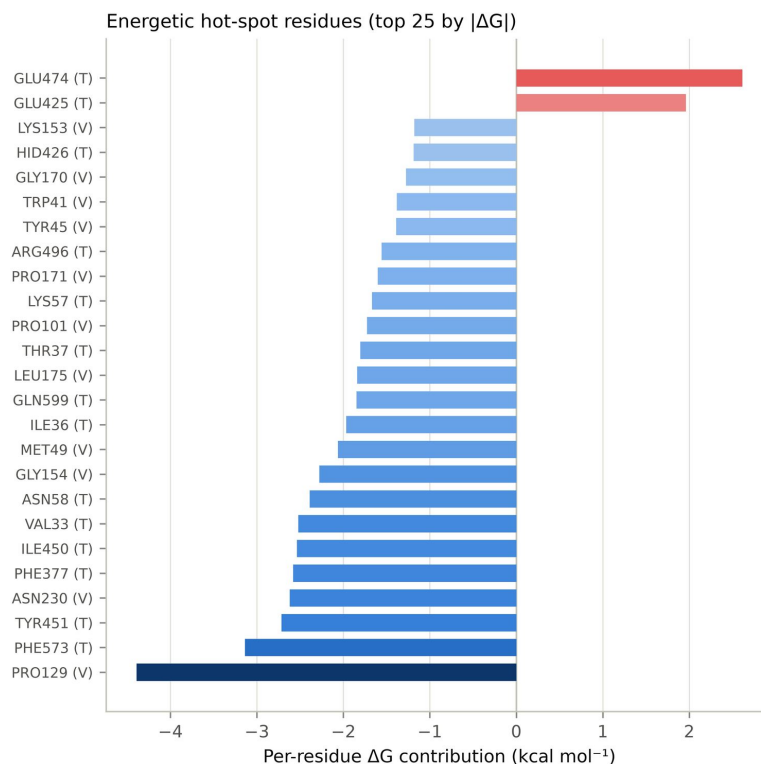
MM/PBSA binding free energy GB -68.85 · PB -108.97 kcal mol⁻¹



Decomposed binding free energy under both solvation models. Van der Waals and non-polar solvation drive binding, opposed by the polar desolvation penalty. Net binding free energy: -68.9 kcal/mol (Generalised Born) and -109.0 kcal/mol (Poisson-Boltzmann).

Figure 13. MM/GBSA convergence

Running average of the GB binding energy against frame count. The estimate is stable over the final third of the trajectory, confirming that sampling is sufficient for the reported value.

Figure 14. Energetic hot-spot residues

Per-residue free energy decomposition, top 25 by magnitude. PRO129, PHE573 and TYR451 dominate the favourable contribution; GLU474 and GLU425 are net unfavourable, marking them as candidates for affinity-maturation mutagenesis.

What a full project delivers

- **High-resolution figures** 300-600 dpi raster, prepared to journal specification
- **Vector graphics** SVG, PDF and EPS versions of every panel
- **Raw trajectory files** .xtc and .trr, plus PBC-corrected and fitted trajectories
- **Topology & parameters** Complete system definition so the run can be reproduced
- **Analysis scripts** Every figure regenerable from the code shipped with the report
- **Data tables** CSV and Excel exports of the numbers behind each curve
- **Statistical summaries** Means, standard deviations, block averaging, convergence tests
- **Interpretation report** A written 2-10 page scientific analysis, not just plots
- **Figure legends** Drafted in journal style, ready to paste into your manuscript
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